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Characteristics of the soil macropore and root architecture of alpine meadows during the seasonal freezing-thawing process and their impact on water transport in the Qinghai Lake watershed, northeastern Qinghai–Tibet Plateau

X. Hu, L.-B. Jiang, Y.-D. Zhao, and X.-Y. Li

Abstract: Low temperatures, freezing-thawing cycles, and short growing seasons characterize alpine soils. The matic epipedon, a special diagnostic surface horizon with an intensive root network, is widely distributed in alpine ecosystems. Studies on the soil macropores and roots of the matic epipedon layer in response to seasonal freezing-thawing processes on the Qinghai–Tibet Plateau are lacking. This study characterized the soil macropores and roots of alpine meadows during the seasonal freezing-thawing process using X-ray computed tomography and revealed the influence of soil macropores and roots structure on water transport in the matic epipedon layer of the alpine meadows. The results showed that the soil pore distribution was more uniform during the unstable freezing stage (UFP) and the unstable thawing stage (UTP), whereas there was a clear matic epipedon layer during the completely thawed stage (TP) and the completely frozen stage (FP). Soils in the TP stage had a higher total surface area density ($0.1898 \text{ mm}^2 \text{ mm}^{-3}$), length density ($225.28 \text{ mm mm}^{-3}$), node density ($1,592 \text{ no. mm}^{-3}$), and connectivity (0.3144) of soil macropores than those in the UFP, UTP, and FP stages. In the TP stages, the density, surface area density, branch density, length density, and node density of roots had significant correlations with the macroporosity, surface area density, and tortuosity of soil macropores. In the FP stage, there were no correlations between the root and soil macropore characteristics. Vertical water is expected to move more readily through the matic epipedon in the TP stage than in the UFP and UTP stages. Roots were the preferential pathways for water transport into the soil layer of the alpine meadow. Therefore, the matic epipedon is a key layer for water and nutrient storage and plays an important role in the water-holding function of the Tibetan Plateau due to its greater root development.

Key words: matic epipedon—root—seasonal freezing-thawing—soil macropore—soil water

Low temperatures, freezing-thawing cycles, and short growing seasons characterize alpine soils. Alpine meadows cover an area of $450,000 \text{ km}^2$ in China (Institute of Geography 1990). The matic epipedon is widely distributed in alpine ecosystems (Zhi et al. 2017). The matic epipedon is a rich surface layer with intertwined grass roots,

which has been recognized in Chinese soil taxonomy as “matic epipedon” (Gong 1999; Kaiser et al. 2008; Zeng et al. 2013). The matic epipedon has an intensive root network that protects soils against trampling by livestock; in addition, its massive underground biomass absorbs and stores a large amount of nutrients to support rapid regrowth of roots

after grazing (Miehe et al. 2011; Schmitt et al. 2013; Sun et al. 2018). Many studies have indicated that the matic epipedon in alpine ecosystems plays an important role in soil water retention due to its densely intertwined grass roots and its high organic matter content (Zhang et al. 2015; Zhi et al. 2017). Hu et al. (2015) reported that the large number of macropores in the soil surface layer of alpine meadows could be attributed to a matic epipedon. To date, research on soil macropore and root structure responses to seasonal freezing-thawing processes on the Qinghai–Tibet Plateau is still lacking.

Soil structure is the three-dimensional (3-D) arrangement of solid constituents and voids at different scales (Rabot et al. 2018), resulting from interactions of biotic and abiotic factors including climate, mineral composition, organic matter, roots, fungal hyphae, soil fauna, and tillage. Soil structure regulates the soil's water retention and infiltration, gas exchange, organic matter and nutrient dynamics, root penetration, and susceptibility to erosion (Jarvis 2007). Soil structure also provides habitat for a myriad of soil organisms, consequently driving their diversity and regulating their activity (Elliott and Coleman 1988; Bach et al. 2018; Xia et al. 2022). X-ray computed tomography (CT) has been used as a new method to characterize soil macropores and roots (Katuwal et al. 2015; Hu et al. 2022; Zhao et al. 2021). Hu et al. (2022) identified water transport through soil pores or roots using CT in shrub-encroached grassland. The above studies indicated that CT is a suitable method to characterize the roots and soil macropores in soil column.

The freezing-thawing cycle (FTCs) has a great impact on soil pore through, for example, reducing the size of soil aggregates (Starkloff et al. 2017), increasing soil porosity (Gao et al. 2020; Zhao et al. 2021), and increasing soil water transport (Ishikawa et al. 2016). Zhao et al. (2021) indicated that soil porosity showed an increase-decrease varia-

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tion trend during successive FTCs, and the connectivity of soil pores and soil water infiltration rate decreased considerably after 10 FTCs. Leuther and Schlüter (2021) showed that 2 to 5 FTCs significantly affected the soil structure and that the effects per FTC on the tested parameters decreased with the number of FTCs. However, it is unknown whether seasonal changes increase or decrease the soil porosity and root density in alpine meadows, and whether the changes in the soil macropores and roots of the mattic epipedon layer due to seasonal FTCs affect the distribution of the soil moisture in these meadows.

We hypothesized that (1) soil macropore and root characteristics were significantly different among seasonal freezing–thawing stages; (2) soils in the completely thawed stage (TP) had a higher total surface area density, length density, node density, and connectivity of soil macropores than those in the completely frozen stage (FP), the unstable freezing stage (UFP), and the unstable thawing stage (UTP); and that (3) roots were the preferential pathways of water transport into the deep soil depth in the alpine meadow. So, the objectives of this study were (1) to quantify soil macropores and roots of alpine meadows during the seasonal freezing–thawing process using CT scanning, and (2) to reveal the influence of soil macropores and roots on the water transport in the mattic epipedon layer of alpine meadows.

Materials and Methods

Study Sites and Core Samples. This study was carried out in the Qinghai Lake watershed on the northeastern Qinghai–Tibet Plateau (figure 1). The alpine meadow ecosystem (37°27′01.46″ N, 100°16′10.07″ E) was selected in the study area, with an altitude of 3,600 m. According to the soil temperature of the alpine meadow obtained by four soil temperature probes (Campbell Scientific 109, Logan, Utah) at 0.5 Hz and with 30-minute averages (figure 2), the seasonal freeze–thaw process was divided into four periods in this study: the TP stage (soil temperature completely above zero), the UFP stage (soil temperature starts to go to zero), the FP stage (soil temperature completely below zero), and UTP stage (soil temperature starts above zero).

Twelve soil columns were collected during seasonal FTCs, with three replicates per period (Hu et al. 2015, 2016). Polyvinyl chloride (PVC) cylinders (7.5 cm diameter,

Figure 1
The location of sampling site.

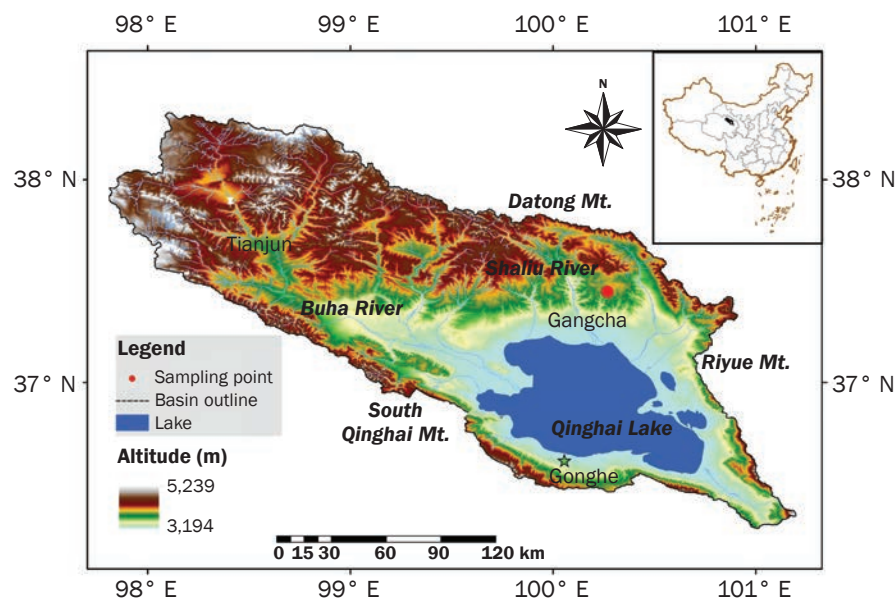
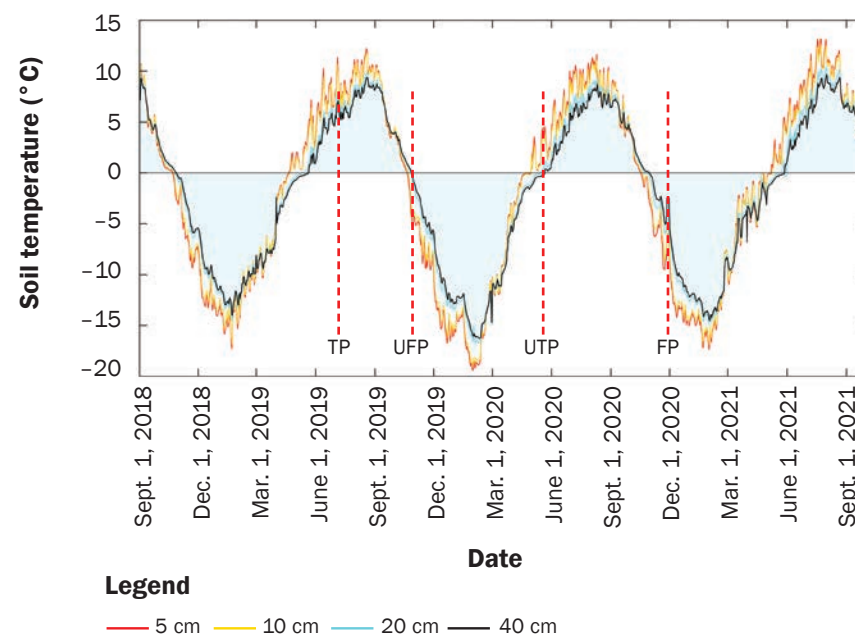


Figure 2

Soil temperature of the study site in 2018 to 2021. UTP is unstable thawing stage; FP is completely frozen stage; UFP is unstable freezing stage; and TP is completely thawed stage.



0.4 cm thick walls, and 30 cm length) were used to collect soil cores. Core extraction followed the previously described procedure (Hu et al. 2015, 2016). In the TP, UFP, FP, and UTP stages, soil cores were extracted

by an excavator. Soil cores were stored in ice boxes and transferred to the laboratory within 24 hours.

The soil particle-size distribution was measured by a MasterSizer 2000 (Malvern

Panalytical Ltd.), and the particle size composition was classified into clay (<2 µm), silt (2 to 50 µm), and sand (50 to 2,000 µm) particles according to the US grading standard. Soil organic carbon (SOC) was measured by a C nitrogen (N) analyzer (CN802, VELP, Italy). The soil characteristics of different soil types are listed in table 1.

Computed Tomography Scanning. A Geoscan-200 X-ray micro-CT (Sanying Precision Instruments Company, Tianjin, China) with a voltage of 150 kV and a current of 250 µA was used to scan intact soil cores after each FTC. After each CT scan, low-noise images with 80 µm resolutions were obtained for each soil cores.

Soil Macropore Analysis. The quantification of soil macropores followed the procedure of Hu et al. (2022). The threshold of soil macropores was determined according to the greyscale value (figure 3). The threshold value was put into all slices to obtain binarization data. After skeletonization, all soil macropore parameters were quantified (Hu et al. 2019).

After skeletonization, macropore number density (g_d [m⁻³]) was defined as the number of macropore networks (N_m [–]) in a unit volume (V [m³]) (equation 1):

$$g_d = \frac{N_m}{V} \quad (1)$$

A node refers to an intersection at which two macropore branches are connected. The number of nodes in each macropore network was recorded, and we calculated the node density (ρ_n [m⁻³]) as the number of nodes (N_n [–]) in a unit volume (V [m³]) (equation 2):

$$\rho_n = \frac{N_n}{V} \quad (2)$$

The length density (g_l [km m⁻³]) was also analyzed as the total actual macropore length (L_t) in a unit volume (V) (Luo et al. 2010) (equation 3):

$$g_l = \frac{L_t}{V} \quad (3)$$

One macropore network may consist of several branches of connected macropores or just one individual macropore. The branch density is the number of branches in a unit volume. The branch density was calculated as equation 4:

$$\rho_b = \frac{n}{V} \quad (4)$$

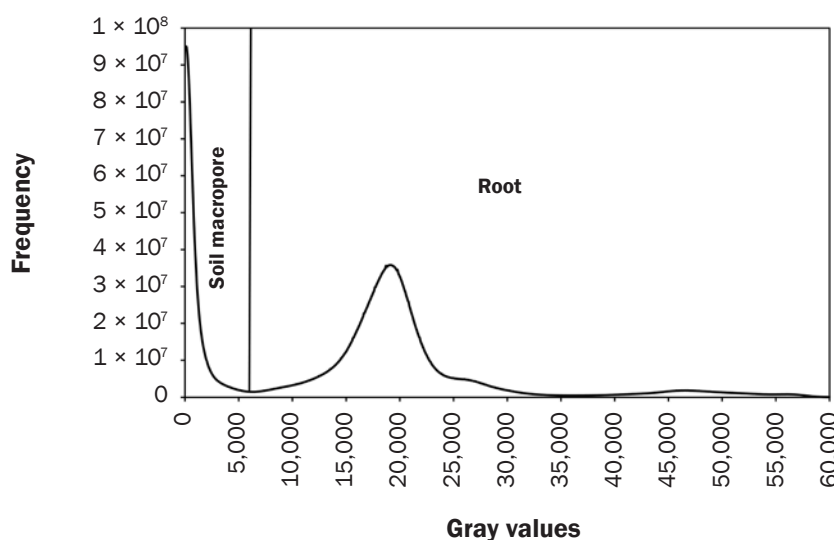
Table 1

Physical and chemical properties of soil in the study area.

Soil depth (cm)	Bulk density (g cm ⁻³)	SOC (g kg ⁻¹)	pH	Clay (%)	Silt (%)	Sand (%)
0 to 10	0.95	84.25	5.80	5.40	47.82	46.78
10 to 20	1.07	42.24	5.98	5.05	48.91	46.04
20 to 30	1.33	6.44	6.38	4.81	46.11	49.08

Figure 3

Gray values of soil columns scanned using X-ray computed tomography (CT).



The mean tortuosity (τ) was calculated as ratio of the total actual macropore length (L_t) to total straight-line distance (L_n) of all the macropores in a certain volume (Luo et al. 2010). Also, the mean tortuosity means the ratio of total pore length to that of the shortest possible path (equation 5):

$$\tau = \frac{\sum_{i=1}^n L_{ti}}{\sum_{i=1}^n L_{li}} \quad (5)$$

where i is the index of a macropore branch and n is the total number of macropore branches.

The inclination of a macropore branch was characterized by an angle (θ) away from vertical direction. The mean angle was calculated as equation 6:

$$\bar{\theta} = \frac{\sum_{i=1}^n L_{ti} \theta_i}{\sum_{i=1}^n L_{li}} \quad (6)$$

Assuming that all macropores are cylindrical, its mean equivalent hydraulic radius was calculated by the total volume (V_t) and total length (L_t) of the macropore (equation 7):

$$\bar{r} = \sqrt{\frac{V_t}{L_t \pi}} \quad (7)$$

Root Segmentation and Analysis. Roots were interpreted by a combination of the growth method and the threshold method (Hu et al. 2018; Kuka et al. 2013). A range of threshold value (5,100 to 60,000) was selected according to the greyscale histogram (figure 3). The magic wand tool in Avizo 9.0 (FEI 2016) was then used to select the roots in the 2-D images, and the root morphology was displayed in the window after application to all slices. It was necessary to enlarge the selected area to ensure that the valid inner voxels were counted as roots. If the selected root area exceeded the actual parts, all voxels were shrunk until the selected root area was reasonable. All the above procedures were done manually using Avizo 9.0.

Water Transport in the Soil. In the TP stage, a dye tracing method was used to measure soil water transport (Peng et al. 2013).

Three plots of 50 × 50 cm were set up on the sample plot, and the plots were all enclosed with 50 × 50 cm tin frames before staining, and the 40 cm high tin was buried 30 cm into the ground, which could prevent the lateral infiltration of water to a certain extent. Twenty-five, 50, and 75 L Brilliant Blue solution with a concentration of 3.0 g L⁻¹ was used as a tracer, and the solution was stirred well and slowly poured into the plots. After 24 hours of infiltration process, the stained area was excavated, and the maximum staining depth of the profile was measured with a tape measure and the stained profile was photographed with a digital camera. Avizo 9 software was used to analyze the depth of staining, the percentage of staining, the staining paths, and the width of the staining paths in the different occurrence layers of the soil.

Statistical Analyses. All the statistical analyses except the redundancy analysis (RDA) were conducted with IBM's SPSS 20 software (SPSS Inc., Chicago, Illinois). One-way analysis of variance (ANOVA) was conducted to compare the differences between four seasonal freezing-thawing stages. Pearson's correlations were used to quantify the linkages among the water transport, soil macropores, and roots during the seasonal freezing-thawing stages.

Results and Discussion

Soil Macropore Network Characteristics. 3-D visualization of the soil macropores in the alpine meadow during the seasonal freezing-thawing process is showed in figure 4. The soil pore distribution was more uniform during the UTP stage and the UFP stage, whereas there was a clear matic epipedon layer during the FP stage and the TP stage.

The characteristics of soil macropore during the seasonal freeze-thaw process are shown in table 2 and figure 5. The mean macroporosity at the TP stage was the largest (0.0214 μm³ μm⁻³), exceeding the mean macroporosity at the FP stage (0.0163 μm³ μm⁻³), the UTP stage (0.0075 μm³ μm⁻³), and the UFP stage (0.0057 μm³ μm⁻³) (table 2). Soils in the TP stage had the large surface area density (0.1898 mm² mm⁻³), length density (225.28 mm mm⁻³), and node density (1.952 × 10³ mm⁻³), whereas soils in the FP stage had a low total surface area density (0.1326 mm² mm⁻³), length density (390.89 mm mm⁻³), and node density (1.45 × 10³ mm⁻³). Soils in the UTP stage had a greater surface area density (0.088 mm² mm⁻³), length density (76.36 mm mm⁻³), and node density (0.976 × 10³ mm⁻³) than soils in the UFP stage (0.0608 mm² mm⁻³, 56.83 mm mm⁻³, and 0.614 × 10³ mm⁻³). There were signifi-

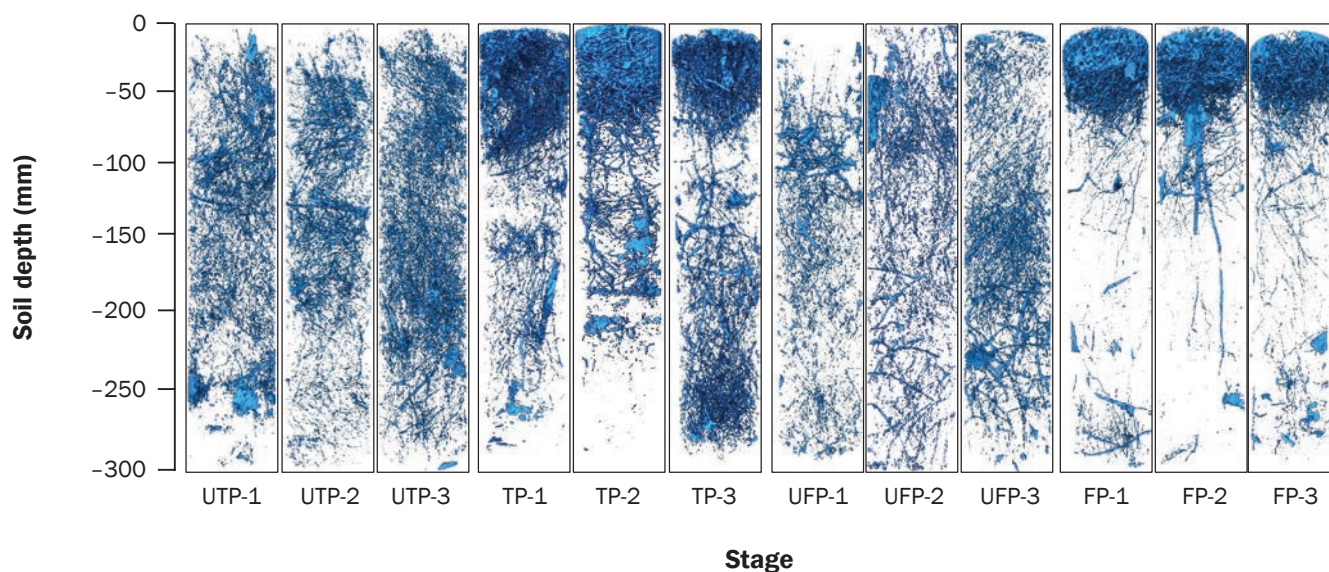
cant differences in soil macroporosity, surface area density, and length density between the UFP/FP and UTP/TP stages.

The mean tortuosities were 1.053, 3.294, 1.329, and 6.735 for the UTP stage, TP stage, UFP stage, and FP stage (table 2). The well-connected macropores in the FP stage and TP stage had mean connectivities of 0.7268 and 0.3144, respectively, while those in the UFP stage and the UTP stage were 0.0561 and 0.0164, respectively. The tortuosities and connectivities of soil macropores in the TP stage and the FP stage were higher than those in the UFP stage and UTP stage. Significant differences were observed in connectivity and tortuosity of soil macropore between the UFP/TP and UTP/FP stages.

Soils in the UTP stage and the TP stage had mean angles of 58.23° and 56.55°, respectively, which indicated that the macropores were predominantly longitudinally oriented. Soils in the UFP stage and the FP stage had relatively small mean angles of 53.70° and 51.69°. The mean branch density in the TP stage was 2,958 no. mm⁻³, which was approximately two times that in the UTP stage (1,146 no. mm⁻³) (table 1). The macropore branch density in the FP stage was 2,717 no. mm⁻³, which was higher than that in the UFP stage (797 no. mm⁻³).

Figure 4

3-D visualization of soil macropores in an alpine meadow during the seasonal freezing-thawing process. UTP is unstable thawing stage; FP is completely frozen stage; UFP is unstable freezing stage; and TP is completely thawed stage. Blue denotes soil macropores.



Similarly, the macropore number densities in the FP stage (1.656×10^3 no. mm^{-3}) and the UTP stage (1.512×10^3 no. mm^{-3}) were larger than those in the UFP stage (0.942×10^3 no. mm^{-3}) and the TP stage (1.448×10^3 no. mm^{-3}).

Root Network Characteristics. 3-D visualizations of roots during the seasonal freezing-thawing process are shown in figure 6. Root characteristics clearly differed among seasonal freezing-thawing processes. Roots were dominated in the surface layer in the FP stage and the TP stage.

The root characteristics during the seasonal freezing-thawing process are showed in table 3. The root density in the FP stage was the high ($0.0415 \mu\text{m}^3 \mu\text{m}^{-3}$), followed by those in the TP stage ($0.0204 \mu\text{m}^3 \mu\text{m}^{-3}$) (table 3). Significant differences were found in the root density between the FP stage and the TP stage.

The root surface area density in the FP stage ($0.2576 \text{ mm}^2 \text{ mm}^{-3}$) was greater than the density in the TP stage ($0.1634 \text{ mm}^2 \text{ mm}^{-3}$). The mean root length density and node density in the FP stage and the TP stage were $0.417 \times 10^3 \text{ mm mm}^{-3}$ and $2.597 \times 10^3 \text{ mm}^{-3}$, respectively. The root branch densities in the TP stage and the FP stage were $4.867 \times 10^3 \text{ mm}^{-3}$ and $2.247 \times 10^3 \text{ mm}^{-3}$, respectively (table 3). There were significant differences in the mean root density between the FP stage and the TP stage.

Relationship between Roots, Soil Macropores, and Water Transport. Tables 4 and 5 show the relationships between the

Figure 5

Distribution of computed tomography (CT)-measured macroporosity along soil column depths during the seasonal freezing-thawing process. UTP is unstable thawing stage; FP is completely frozen stage; UFP is unstable freezing stage; and TP is completely thawed stage.

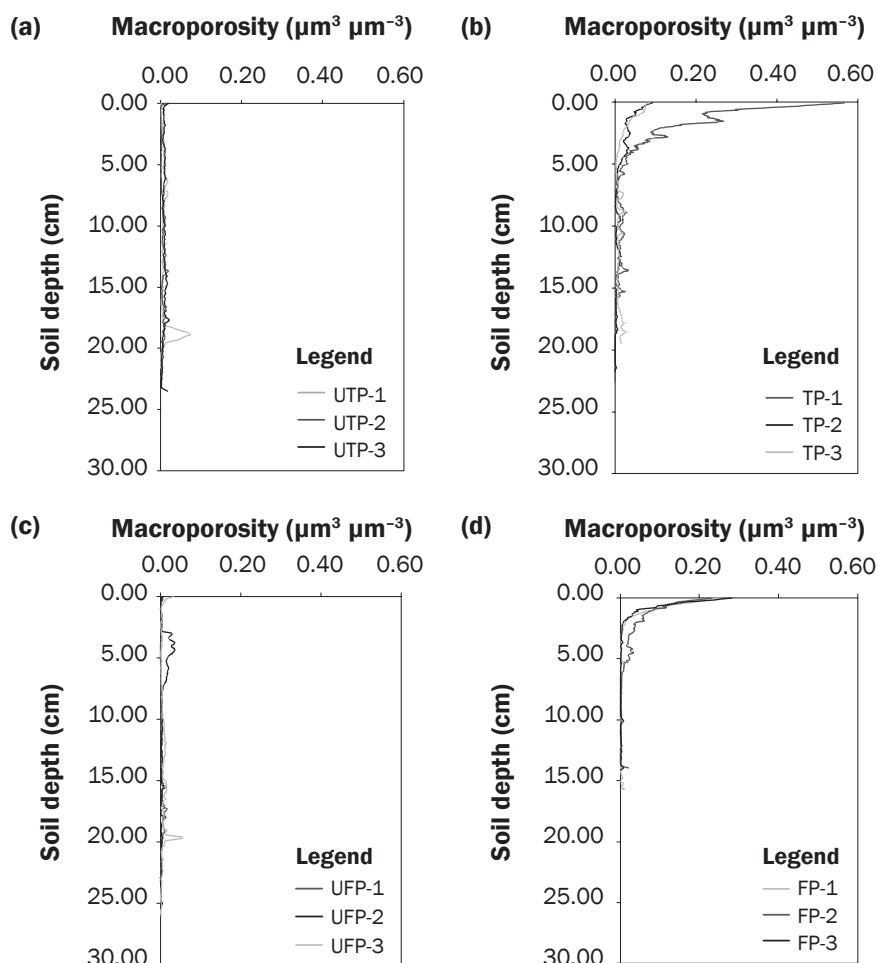


Table 2

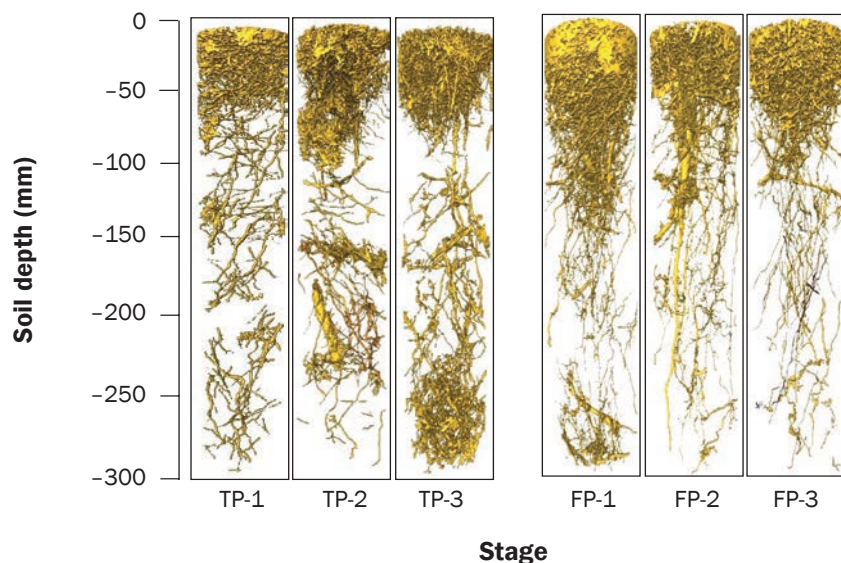
Soil macropore characteristics of all soil columns as determined by X-ray computed tomography (CT) during the seasonal freeze-thaw process.

Stage	Macro-porosity ($\mu\text{m}^3 \mu\text{m}^{-3}$)	Mean diameter (mm)	Mean volume (mm^3)	Mean angle ($^\circ$)	Surface area density ($\text{mm}^2 \text{mm}^{-3}$)	Number density (no. mm^{-3})	Branch density (no. mm^{-3})	Length density (mm mm^{-3})	Node density (no. mm^{-3})	Hydr-aulic radius (mm)	Conn-ectivity	Tortuosity (-)
UTP	$0.0075 \pm 0.0016a$	$0.2152 \pm 0.0005a$	$0.0510 \pm 0.0124a$	$58.23 \pm 1.02a$	$0.0880 \pm 0.0174a$	$1,512 \pm 371a$	$1,146 \pm 252ab$	$76.36 \pm 14.55a$	$976 \pm 217a$	$0.1765 \pm 0.0149a$	$0.0164 \pm 0.0196a$	$1.0527 \pm 0.0797a$
TP	$0.0214 \pm 0.0130b$	$0.2080 \pm 0.0225a$	$0.1395 \pm 0.0652b$	$56.55 \pm 1.65ab$	$0.1898 \pm 0.0819b$	$1,448 \pm 220a$	$2,958 \pm 1,872bc$	$225.28 \pm 140.45b$	$1,952 \pm 872a$	$0.1749 \pm 0.0018a$	$0.3144 \pm 0.2537b$	$3.2937 \pm 2.2106b$
UFP	$0.0057 \pm 0.0016a$	$0.2120 \pm 0.0197a$	$0.0602 \pm 0.0151a$	$53.70 \pm 1.74b$	$0.0608 \pm 0.0097a$	$942 \pm 78b$	$797 \pm 104a$	$56.83 \pm 6.69c$	$614 \pm 109b$	$0.1770 \pm 0.0268ab$	$0.0561 \pm 0.0657ab$	$1.3296 \pm 0.1817a$
FP	$0.0163 \pm 0.0042b$	$0.1567 \pm 0.0033b$	$0.0990 \pm 0.0031b$	$51.69 \pm 11.61ab$	$0.1326 \pm 0.0225b$	$1,656 \pm 441.1347a$	$2,717 \pm 327c$	$390.89 \pm 220.79b$	$1,459 \pm 237c$	$0.1237 \pm 0.0178b$	$0.7268 \pm 0.0121c$	$6.7348 \pm 1.9111b$

Notes: Column marked by the same letter is not significantly different at $p = 0.05$ between the treatments; the absence of letters indicates no significant differences. Data in the parentheses are standard deviations. UTP = unstable thawing stage. FP = completely frozen stage. UFP = unstable freezing stage. TP = completely thawed stage.

Figure 6

3-D visualization of root networks during the seasonal freezing-thawing process. FP is completely frozen stage and TP is completely thawed stage. Yellow denotes roots.



roots and soil macropores during the freezing-thawing stages. In the FP stage, there were no correlations between the roots and soil macropore characteristics. In the TP stage, the density, surface area density, branch density and length density of roots had significant positive correlation with macroporosity, surface area density, number density, branch density, and node density of macropore. The density, surface area density, branch density, and length density of roots had significant negative correlation with hydraulic radius of macropore.

Figure 7 shows the changes in dyed areas with soil depth with 25, 50, and 75 L Brilliant Blue solution. Correlations between the soil macropore, root, and the dyed area ratio are shown in table 6. The results showed that the dyed area ratio had significant relationships

with root parameters and soil macropore parameters. The dyed area ratios with 25 L Brilliant Blue solution had significant positive correlation with the surface area density, branch density, length density, node density, and tortuosity of soil macropores and roots ($p < 0.05$). There was significant and negative correlation between the root equivalent diameter and the dyed area ratios with 50 L Brilliant Blue solution. There were significant and positive correlations between the dyed area ratios with 75 L Brilliant Blue solution and the mean volume of macropores and mean angle of roots ($p < 0.05$). There were significant and negative correlations between the dyed area ratios with 75 L Brilliant Blue solution and root equivalent diameter and root branch length.

Discussion. The results showed that the soil pore distribution was more uniform during the UTP stage and the UFP stage, whereas there was a clear matic epipedon layer during the FP stage and the TP stage. Soil temperature fluctuates near 0°C several times in a short period of time during the UTP stage and the UFP stage, and the soil moisture causes the soil to freeze and thaw and shrink (Wang et al. 2016). Soil macroporosity of the UFP stage was reduced by 73% compared to the FP stage, and soil macroporosity of the UTP stage decreased by 54% compared to the TP stage (figure 5). So, soil macroporosity was decreased during FTCs. This is consistent with other results. Gao et al. (2021) showed that the soil macroporosity was reduced by about 57% during FTCs. FTCs can affect soil structure differently depending on soil texture (Angin et al. 2016; Fouli et al. 2013; Kværnø and Øygarden 2006; Oztas and Fayetorbay 2003). Zhang et al. (2016) pointed out that the soil structure after FTCs can change significantly due to the fragmentation of coarse particles and the aggregation of fine particles. Zhao et al. (2021) indicated that freeze-thaw could lead to the fragmentation of silt and sand, which determined the effect of FTCs on soil pores. So, coarse particles would be dispersed (broken) and the fine particles would be aggregated during FTCs. Correspondingly, in the FP stage and the TP stage, soil macroporosity was increased at soil layer of 0 to 10 cm.

In the same soil samples, Zhao et al. (2021) indicated that soil total porosity (0 to 280 mm) was prevailed by soil porosity in the 0 to 75 mm soil layers (76.3% to 91.4%), which led to an increase in the soil layer (0 to 75 mm) after 10 FTCs (table 7). During FTCs, soil porosity decreased after 10 FTCs

Table 3

Root characteristics of all soil columns as determined by computed tomography (CT) images during the seasonal freeze-thaw process.

Stage	Root density	Mean angle (°)	Surface area density (mm ² mm ⁻³)	Branch density (no. mm ⁻³)	Length density (mm ² mm ⁻³)	Node density (no. mm ⁻³)	Mean diameter (mm)	Mean branch length (mm)	Tortuosity (-)
TP	0.0204 ± 0.0039b	56.93 ± 1.07	0.1634 ± 0.0317bc	2,247 ± 446b	172.24 ± 35.81b	1,318 ± 241b	0.0235 ± 0.0003a	0.7656 ± 0.0210	51.68 ± 35.57ab
FP	0.0415 ± 0.0125c	54.98 ± 4.60	0.2576 ± 0.0760c	4,867 ± 1,919c	416.60 ± 109.71c	2,597 ± 1,046c	0.0216 ± 0.0010b	0.9026 ± 0.1587	119.76 ± 77.91b

Notes: Column marked by the same letter is not significantly different at $p = 0.05$ between the treatments; the absence of letters indicates no significant differences. Data in the parentheses are standard deviations. FP = completely frozen stage. TP = completely thawed stage.

Table 4

Correlations between root characteristics and soil macropore characteristics at the completely frozen stage (FP).

Root characteristic	Macro-porosity	Mean diameter	Mean volume	Mean angle	Surface area density	Number density	Branch density	Length density	Node density	Hydraulic radius	Connectivity	Tortuosity
Root density	0.61	-0.73	0.66	-0.03	0.63	0.58	0.64	0.45	0.64	-0.27	0.65	0.53
Root mean angle	0.34	-0.70	0.64	0.72	0.43	0.27	0.49	0.04	0.44	-0.15	0.64	0.28
Root surface area density	0.24	-0.25	0.42	0.43	0.31	0.22	0.34	0.10	0.30	-0.21	0.43	0.24
Root branch density	0.25	-0.27	0.43	0.43	0.32	0.23	0.35	0.11	0.31	-0.23	0.44	0.25
Root length density	0.25	-0.27	0.43	0.43	0.32	0.22	0.35	0.11	0.31	-0.22	0.44	0.25
Root node density	0.25	-0.27	0.43	0.43	0.32	0.22	0.35	0.11	0.31	-0.22	0.44	0.25
Root mean diameter	-0.48	0.36	-0.36	0.23	-0.45	-0.47	-0.43	-0.52	-0.45	0.50	-0.36	-0.51
Root mean branch length	-0.03	0.50	-0.36	-0.76	-0.12	0.03	-0.19	0.23	-0.13	-0.03	-0.36	0.01
Root tortuosity	0.46	-0.69	0.65	0.32	0.52	0.42	0.56	0.21	0.53	-0.06	0.64	0.36

*Correlation is significant at the 0.05 probability level (two-tailed).

**Correlation is significant at the 0.01 probability level (two-tailed).

Table 5

Correlations between root characteristics and soil macropore characteristics at the completely thawed stage (TP).

Root characteristic	Macro-porosity	Mean diameter	Mean volume	Mean angle	Surface area density	Number density	Branch density	Length density	Node density	Hydraulic radius	Connectivity	Tortuosity
Root density	0.88*	-0.96**	0.49	0.15	0.93**	0.98**	0.89*	0.89*	0.93**	-0.90*	0.81	0.82*
Root mean angle	0.28	-0.01	0.15	0.88*	0.29	0.34	0.27	0.27	0.30	-0.18	0.28	0.20
Root surface area density	0.86*	-0.95**	0.47	0.13	0.92**	0.98**	0.88*	0.88*	0.92**	-0.91*	0.80	0.80
Root branch density	0.82*	-0.94**	0.44	0.08	0.88*	0.96**	0.84*	0.84*	0.89*	-0.93**	0.79	0.76
Root length density	0.84*	-0.95**	0.47	0.09	0.90*	0.96**	0.86*	0.86*	0.91*	-0.91*	0.81*	0.79
Root node density	0.80	-0.93**	0.41	0.08	0.87*	0.96**	0.82*	0.82*	0.88*	-0.94**	0.77	0.74
Root mean diameter	-0.59	0.78	-0.12	-0.07	-0.70	-0.90*	-0.62	-0.62	-0.71	0.99**	-0.56	-0.50
Root mean branch length	-0.49	0.58	0.03	-0.33	-0.59	-0.82*	-0.50	-0.50	-0.59	0.89*	-0.39	-0.36
Root tortuosity	0.01	-0.36	-0.31	-0.49	0.10	0.32	0.04	0.04	0.12	-0.67	0.09	-0.03

*Correlation is significant at the 0.05 probability level (two-tailed).

**Correlation is significant at the 0.01 probability level (two-tailed).

in the 75 to 150 mm soil layer, and gradually decreased in the 150 to 280 mm soil layer after 10 FTCs (Zhao et al. 2021). So, soil total porosity increased at the soil depth of 0 to 75 mm, and there was a clear matic epipedon layer in the TP stage and the FP stage after the seasonal freezing-thawing process.

The results showed that the soil macropores and roots were significantly different among the four freeze-thaw stages. In the TP stage, volume density, surface area density, branch density, and length density of roots had significant positive correlation with macroporosity, surface area density, length density, node density, and tortuosity of macropore. However, there was no significant correlation between plant roots and soil porosity in the FP stage, which indicated that plant roots in the growing season contributed to an increase in soil porosity. Previous studies

have shown that the volume density of roots increased with increasing soil porosity (Hu et al. 2020). Matic epipedons of alpine meadows have a dense structure and an abundance of living and dead roots (Kaiser et al. 2008), resulting in a large number of macropores in soil (Yang et al. 2014; Hu et al. 2016). The dispersion and aggregation of soil particles caused by the phase changes of water in the soil resulted in a decreased soil pore volume, surface area, and total porosity. The dense root system of matic epipedons helps them accumulate water (Hu et al. 2019; Stewart et al. 1999). Therefore, the characteristic parameters of roots are negatively correlated with soil porosity in the TP stage.

The results indicated that soils in the TP stage had a larger macropore surface area density, length density, node density, and connectivity than those in the FP stage, the UFP

stage, and the UTP stage. Soil macropores with a high connectivity in alpine meadows soil could increase hydraulic conductivity (Bhattacharyya et al. 2006; Hu et al. 2020). Soils in the UTP stage had a larger macropore surface area density, length density, branch density, and node density than those in the UFP stage. The studies of Borges et al. (2019) and Hu et al. (2020) showed that there was a significant and positive correlation between the soil pore length density and near-saturated water conductivity. Jarvis (2007) found that connected soil macropores accelerated water flow through soil macropore. This is consistent with our findings that soil pore characteristic parameters were significantly positively correlated with the dyed area ratio at a 10 cm water depth. Water is expected to move more readily through the matic epipedon in the TP stage than in the UFP stage

Figure 7

Distribution of dyed area along soil column depths. 25 denotes 25 L Brilliant Blue solution; 50 denotes 50 L Brilliant Blue solution; and 75 denotes 75 L Brilliant Blue solution.

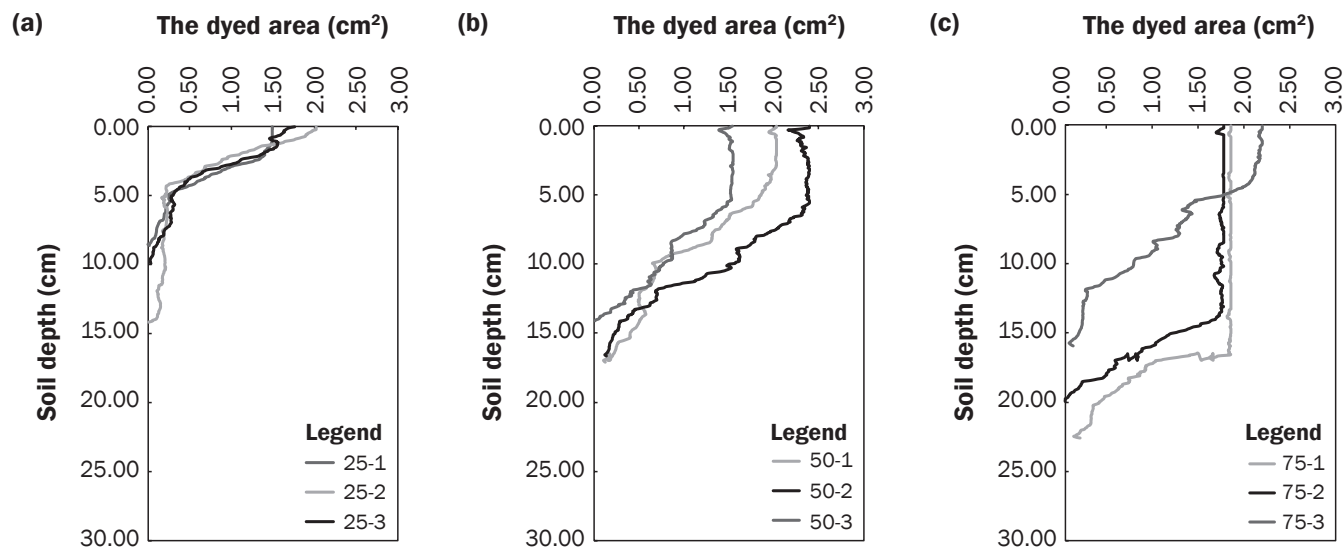


Table 6

Correlations between the dyed area ratios and soil macropore parameters and root parameters.

Parameters		The dyed area ratios		
		25 L KI	50 L KI	75 L KI
Soil macropore parameters	Macroporosity	0.99**	0.69	0.61
	Mean diameter	-0.54	-0.16	0.21
	Mean volume	0.74	0.68	0.83*
	Mean angle	-0.26	0.11	-0.13
	Surface density	0.99**	0.69	0.62
	Number density	0.99**	0.73	0.67
	Branch density	0.98**	0.65	0.57
	Length density	0.99**	0.66	0.58
	Node density	0.99**	0.67	0.59
	Hydraulic radius	-0.6	-0.08	0.15
	Connectivity	0.68	0.2	-0.07
	Tortuosity	0.98**	0.63	0.55
Root parameters	Volume density	0.98**	0.64	0.56
	Mean angle	0.22	0.56	0.84*
	Surface density	0.98**	0.64	0.56
	Branch density	0.98**	0.64	0.55
	Length density	0.98**	0.64	0.55
	Node density	0.98**	0.64	0.56
	Equivalent diameter	-0.93**	-0.86*	-0.85*
	Branch length	-0.5	-0.68	-0.92**
	Tortuosity	0.97**	0.61	0.53

*Correlation is significant at the 0.05 probability level (two-tailed).

**Correlation is significant at the 0.01 probability level (two-tailed).

and the UTP stage (figure 8). Vertical water would move more readily through soil macropore in the UTP stage than those in the UFP stage, and soils in the UFP stage would enhance a greater surface flow than those in the UTP stage accordingly (figure 8).

The results showed that there were significant and positive correlations between the root parameters and the dyed area ratio. Root volume density, surface area density, branch density, length density, and node density are important parameters affecting soil porosity and connectivity. Roots could increase soil porosity, and increased preferential pathways (Feeney et al. 2006; Li et al. 2009). Schwarzel et al. (2012) pointed out that roots induce preferential flows, which triggered rapid lateral subsurface flows. Hu et al. (2015) indicated that the growth and decay of roots resulted in a large number of large pores in the soil, which was conducive to water accumulation. Therefore, the distribution of the root system can affect the soil moisture transport in the alpine meadows.

Summary and Conclusions

The results of this study verified the hypothesis that the soil macropore and root characteristics were significantly different among seasonal freezing-thawing stages. The soil pore distribution was more uniform during the UTP stage and the UFP stage, whereas there was a clear mastic epipedon layer during the TP stage and the FP stage.

Table 7

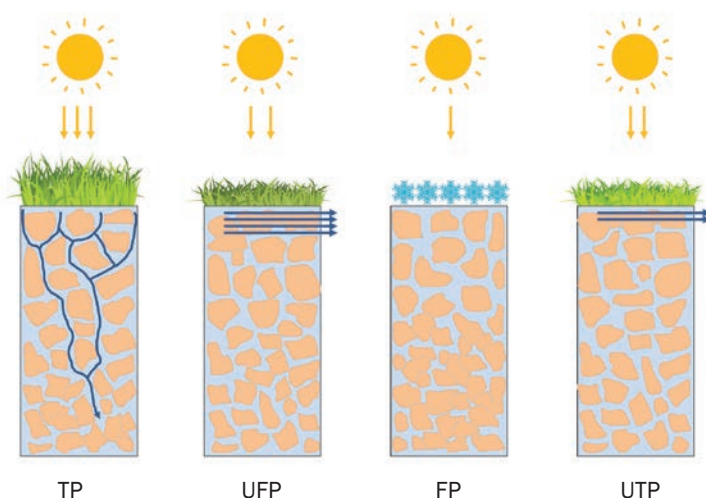
Soil pore characteristics at different soil depths under different freeze-thaw cycles (FTCs) (Zhao and Hu 2021).

Soil depth (mm)	FTCs	Pore number density (10^{-11}) (no. μm^{-3})	Connectivity
0 to 75	0	$0.516 \pm 0.107\text{b}$	$0.097 \pm 0.069\text{a}$
	10	$0.885 \pm 0.160\text{a}$	$0.032 \pm 0.031\text{b}$
75 to 150	0	0.290 ± 0.084	0.064 ± 0.082
	10	0.382 ± 0.184	0.023 ± 0.028
150 to 280	0	0.096 ± 0.081	0.035 ± 0.021
	10	0.043 ± 0.034	0.024 ± 0.003

Notes: Column for each data group by the same letter are not significantly different at $p < 0.05$; the absence of letters indicates no significant differences for that property.

Figure 8

The conceptual model of soil structure dynamics in alpine meadow and its potential mechanisms for its influence on water transport. UTP is unstable thawing stage; FP is completely frozen stage; UFP is unstable freezing stage; and TP is completely thawed stage.



Soils in the TP stage had a larger total surface area density, length density, node density, and connectivity of macropores than those in the FP stage, the UFP stage, and the UTP stage. In the TP stage, the root density, surface area density, branch density, length density, and node density had significant positive correlation with macroporosity, surface area density, length density, node density, and tortuosity of macropores. In the FP stage, there were no correlations between the roots and soil macropore characteristics. Vertical water is expected to move more readily through the matic epipedon in the TP stage than in the UFP stage and the UTP stage. Soils in the UFP stage would facilitate a greater surface flow than those of the UTP stage. Roots were the preferential pathways for moisture

transport into the deep soil depth of the alpine meadow.

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